implosion, with the outward centrifugal forces then fully balancing gravity's pull. Any real star has high density and pressure in its center, and lower density and pressure in its outer layers; as it implodes, high-density lumps will develop here and there like blueberries in a blueberry muffin. Moreover, the star's gaseous matter, as it implodes, will form shock waves —analogues of breaking ocean waves—and these shocks may eject matter and mass from some parts of the star's surface just as an ocean wave can eject droplets of water into the air. Finally, radiation (electromagnetic waves, gravitational waves, neutrinos) will pour out of the star, carrying away mass.

All these effects Oppenheimer and Snyder would have liked to include in their calculations, but to do so was a formidable task, far beyond the capabilities of any physicist or computing machine in 1939. It would not become feasible until the advent of supercomputers in the 1980s. Thus, to make any progress at all, it was necessary to build an idealized model of the imploding star and then compute the predictions of the laws of physics for that model.

Such idealizations were Oppenheimer's forte: When confronted with a horrendously complex situation such as this one, he could discern almost unerringly which phenomena were of crucial importance and which were peripheral.

For the imploding star, one feature was crucial above all others, Oppenheimer believed: gravity as described by Einstein's general relativistic laws. It, and only it, must not be compromised when formulating a calculation that could be done. By contrast, the star's spin and its nonspherical shape could be ignored; they might be crucially important for *some* imploding stars, but for stars that spin slowly, they probably would have no strong effect. Oppenheimer could not really prove this mathematically, but intuitively it seemed clear, and indeed it has turned out to be true. Similarly, his intuition said, the outpouring of radiation was an unimportant detail, as were shock waves and density lumps. Moreover, since (as Oppenheimer and Volkoff had shown) gravity could overwhelm all pressure in massive, dead stars, it seemed safe to pretend (incorrectly, of course) that the imploding star has no internal pressure whatsoever—neither thermal pressure, nor pressure arising from the electrons' or neutrons' claustrophobic degeneracy motions, nor pressure arising from the nuclear force. A real star, with its real pressure, might implode in a different manner